THE RIPPLE EFFECT OF DAILY NEW LOWS

By Ralph Vince and Larry Williams

2024 Charles H. Dow Award Winner

This paper was awarded the
2024 Charles H. Dow Award by the CMT Association.

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Abstract

Breadth data is ubiquitous in the study of the stock market. Daily New Lows have a predictive reliability as robust as any breadth data element, yet remain the most overlooked of all daily breadth metrics, relegated to being the "wallflower" of daily breadth data. Typically, market analysts heavily examine absolute and relative advances/declines and various derivative indicators like the advance-decline line and McClellan Oscillator when assessing daily breadth. Analysts also scrutinize volume on advancing / declining stocks, often combined with advance / decline data in indicators like TRIN.

Herein, we examine the seemingly overlooked predictive reliability of daily New Lows for signaling both the onset and progression of bear markets, the final capitulation phases of declines, and shorter-term interim tops.

Through extensive analysis, we make the case that this reliable, yet underutilized data element warrants daily examination by serious market participants and inclusion in quantitative models - not only in addition to their immediate ramifications, but the persistence of such ramifications into the following weeks and months of significant readings in this mostly-overlooked data point.

The paper aims to demonstrate why daily New Lows merit very serious consideration among the critical pantheon of market breadth indicators.

Introduction

We will be focusing *solely* on *daily* New Lows data of NYSE stocks with respect to price. With any data set, derivative calculations can be formed, examined, and conclusions drawn. For example, moving averages, data differences over various time periods, etc., are all worthy of examination.

We will keep the study of these derivative calculations of daily New Lows outside of the scope of our examination herein, save for the following: as we are dealing with a poisitive, integer value for the number of stocks making new 52-week lows, the number must be put into the context of how many stocks represent the universe of stocks under examination from which the daily number of New Lows is examined. Thus, rather than examine the raw, positive integer value representing the number of daily New Lows of NYSE stocks (Using NYSE as a proxy for "stock markets" in general), we will examine these as a ratio, a percentage of issues traded making daily New Lows on any given day.

We must examine a second restriction to make our examination be not only exhaustive, but formally legitimate. One problem with using NYSE daily New Lows data is the reporting method change which occurred January 2, 1980. Beginning at this time, New Lows were calculated as the lowest price seen of a rolling 52-week lookback. Prior to this time, daily New Lows were calculated for the calendar year, where certain "fudge factor" adjustments where imposed so that, on the first few trading weeks of the year, the previous calendar year was included so as to avoided all stocks making daily New Lows on the first trading day of the year.

Returning to January 2, 1980, we find the total number of issues traded on the NYSE was 1590, and throughout the subsequent months, this number ranged from the high 1400s to the low 1600s. Contrast this to today (late 2023) where the numbers are more than double that of daily issues traded on NYSE. This is the reason for examining our raw data as a percentage of total issues traded. Whereas we wish to use raw "unprocessed" data, this adjustment is necessary given the nature of the data.

Thus, we will be examining the percentage of NYSE stocks making daily, new 52-week lows since January 2, 1980.

We will discuss three very effective uses of this data, both for discerning market bottoms and bear move final phases (the conventionally regarded implementation of this data point) as well as two *novel* applications, both of which provide signals which, peculiarly, induce a persistence to their ramifications into subsequent future months.

Among these, we will detail, a signal for detecting periods of "Extreme Vulnerability", and lastly, another novel signal, one with an exciting track record, which also induces this persistence, but one to tell us when price drops *should* be acted upon defensively.

Review of the Literature

Daily New Lows have been extensively studied as a market indicator with analyses evolving from contrarian sentiment gauge to informing momentum and technical strategies. This review summarizes key publications that have examined the application of daily New Lows in chronological order.

One of the earliest studies on using daily New Lows was posited by Haller[1] wherein he presents looking at various breadth measures versus the broad indexes but also examines the net of New Highs and New Lows. Haller relies on weekly data in his book, but it is the first early mention we find referring to using the aggregate number of annual New Lows.

[1] See Haller, 1965

Haller and Larry Williams-were pen pal friends. According to Williams "Gil was a very precise person who did takes own advice in following the strongest stocks. His background as a former ship captain played a lot into his thinking when we discussed the markets".

Haller's focus on the new Lows/Highs was picked up, chart wise, by Security Market Research a charting service from Denver, Colorado, by using a 10-day moving average of the two values to replicate the Haller work.

A decade after Haller's book, market breadth analysis has well-penetrated the professional stock analysts' offices, as evidenced by Joseph Granville who elaborated at length on market breadth and the effects of daily New Highs and Lows at major market turning points, both in his 1976 book[2] and his previous writings for the famous "Hutton Daily Market Wire."

Another early exposé on New Lows is found in Norman Fosback's 1979 book "Stock Market Logic" which introduced the new high/new low indicator that has since become a widely used technical analysis tool[3]. Fosback found New Lows useful for gauging market psychology and ascertaining potential turning points. The new high/new low indicator quantifies market breadth based on the number of stocks hitting new periodic highs and lows. Fosback laid the groundwork for utilizing New Lows in sentiment analysis.

Alexander Elder's "Trading for a Living" delved deeper into applying daily and weekly New Lows for contrarian analysis[4]. Elder examined New Lows through the lens of crowd psychology, finding extreme spikes in New Lows often marked major market bottoms. He concluded that major shifts in sentiment drove spikes in New Lows at market turning points. Elder advocated combining new low analytics with other sentiment and technical indicators for robust signaling. His work expanded on sentiment applications of New Lows.

In his 1999 book, Gary Smith provided extensive insights into trading strategies utilizing daily New Lows based on his own market experiences[5]. Smith highlighted the crucial importance of tracking New Lows as a sentiment gauge to identify bearish extremes. He covered multiple technical strategies centered around New Lows, while also detailing real-world examples of how he successfully incorporated New Lows analysis into his market forecasts and trades during his career. Smith delivered an in-depth account of applying New Lows in practice.

Charles M.C. Lee and Bhaskaran Swaminathan conducted an empirical investigation into the predictive relationship between New Highs / New Lows and subsequent returns[6].

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[2] See Granville 1976
[3] See Fosback, 1979
[4] See Elder, 1993
[5] See Smith, 1999
[6] See Lee & Swaminathan, 2000

Through quantitative analysis, they established that *increased New Lows reliably predicted higher future returns, while increased New Highs predicted lower future returns*. Their findings formally evidenced New Lows' value as a contrarian sentiment indicator foreshadowing market reversals. The paper contributed robust statistical proof of New Lows' predictive abilities.

Combining behavioral finance and technical analysis, Ronald Anderson, Anup Agrawal and Jeffrey Jaffe examined market extremes through the lens of New Highs and Lows[7]. They found New Lows were often driven by investor overreaction and shifts in sentiment. Moreover, combining New Lows data with complementary technical indicators proved useful for objectively identifying market extremes. The paper highlighted the value of blending New Low analytics with other frameworks.

Demonstrating New Lows' applicability beyond contrarian trading, Thomas George and Chuan-Yang Hwang researched New Highs as a momentum indicator[8]. They established that stocks hitting 52-week highs strongly tended to continue moving higher over the short-term. Their evidence formalized New Highs' ability to signal momentum persistence. The findings suggested incorporating New Highs data could enhance momentum strategies and confirm market strength.

In his technical analysis book, Gerald Appel advocated tracking daily New Lows to gauge bearish sentiment, especially during latter bear market stages[9]. Appel found properly interpreting spikes in New Lows proved useful for identifying capitulation bottoms that marked reversals. He focused especially on contrarian utilities of New Lows for predicting impending trend changes late in bear markets. The work expanded on using New Lows for change in sentiment.

Taking a scholarly approach, Menachem Brenner and Rafi Eldor statistically analyzed the symmetry and persistence of New Highs and Lows in market data[10]. They discovered New Highs and Lows exhibited clear symmetrical relationships, whereby periods of increased new high frequency were followed by spikes in New Lows, and vice-versa. Their findings evidenced recurring cyclicality and mean reversion tendencies between New Highs and Lows. The paper contributed mathematical proof of New Lows' tendencies.

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[7] See Anderson, Agrawal & Jaffe, 2006
[8] See George & Hwang, 2004
[9] See Appel, 2005
[10] Brenner & Eldor, 2009

Discussion

We now examine three methods of utilizing daily NYSE New Lows data for actionable signals in the stock market.

1. The Selling Climax Signal

This is among the oldest used applications of daily New Lows data. Essentially, the idea is that market bottoms naturally witness an excessive number of daily New Lows (as would be expected).

Many types of indicators have been constructed of all sorts of data over the decades to detect this, these so-called "selling climax" situations. Ultimately, any such indicators are ineluctably redundant to simply looking at daily New Lows.

According to Haller, "When 750 or more net New Lows are recorded within a week, that week is the first climax week. Buy at the start of the fourth week after the first climax week."

Examination of the total number of issues in 1965, the year of publication and the few years leading up to it, reveals there were roughly 1000 to the low 1100s of issues trades.

Thus, Haller's 750 or more net New Lows can be translated, roughly, into being "70% of total issues."

Figure 1 shows us, over the time period we are considering (January 2, 1980 through November 20, 2023), the weekly New Lows along with a green horizontal line at this 70% mark of Haller's.

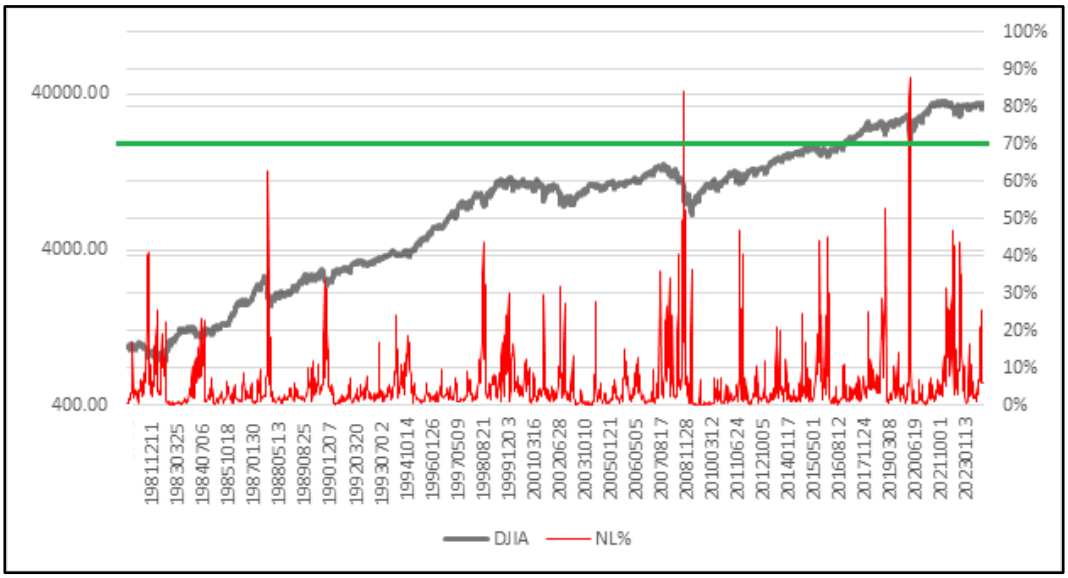


Figure 1: Lognormal Dow Industrials in black and weekly New Lows percentage, weekly since 1980:

There have been two such signals in this nearly 44-year time period:

84% Oct 3, 2008

88% March 13, 2020

The daily difference in calculation of daily New Highs and Lows, post-Haller, may account in the weekly data, perhaps, such that a lower threshold, 50%, perhaps even less, may be more germane to markets in the post-1980 world.

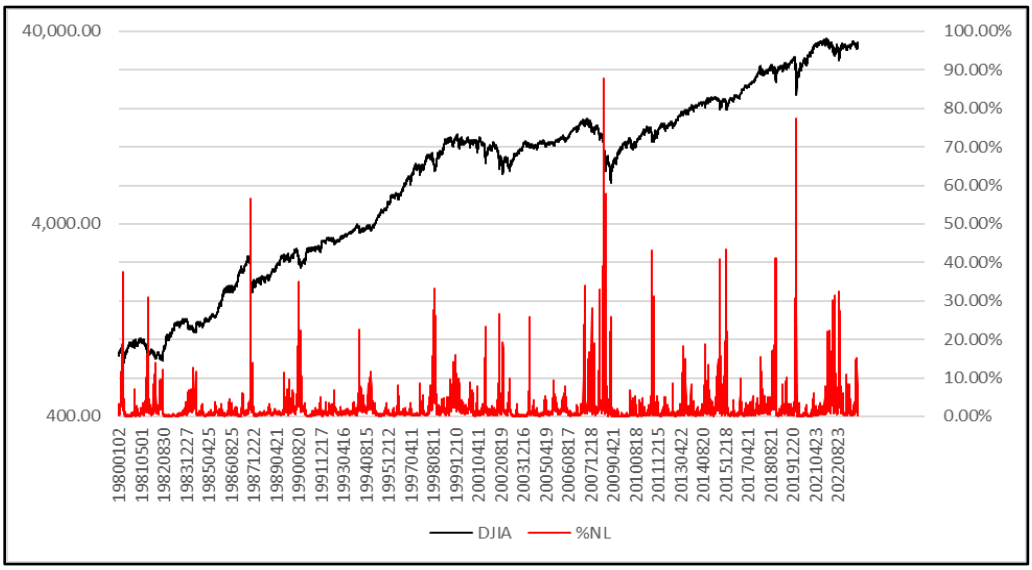


Figure 2: Lognormal Dow Industrials in black and daily New Lows percentage since 1980

Employing daily data, we seek to re-examine Haller's original rule of buying "at the start of the fourth week after a reading of 70% New Lows," amending it for daily data. Our results are displayed in Table 1. As such, we look at various thresholds, 70%, 50%, 40%, 30%, and 20% daily issues making 52-week New Lows, and buy on the close 22 market days after such a reading of New Lows. We will look how such positions play out 22, 65, 130, and 260 market days afterwards to replicate 1, 3, 6 and 12 months after entry. We will use S&P 500 Index data and percentage gains and losses given the disparity of index prices over the near 44-year test period between January 2, 1980 and November 20, 2023.

Table 1. Entering 22 market days after recording daily New Lows in the far left column, exiting the position 22, 65, 130 and 265 days later using S&P 500 prices:

	22	65	130	260	<--Days after Entry to Exit
NL%					
70%	2	2	2	2	Total Trades
	1	1	1	2	Winning Trades
	50.00%	50.00%	50.00%	100.00%	Percent Winners
	1.176%	6.643%	0.259%	70.462%	Cumulative % Gain
	0.588%	3.322%	0.130%	35.231%	Average % Gain
50%	4	3	3	3	Total Trades
	1	2	2	3	Winning Trades
	25.00%	66.67%	66.67%	100.00%	Percent Winners
	-12.199%	12.529%	24.269%	74.616%	Cumulative % Gain
	-3.050%	4.176%	8.090%	24.872%	Average % Gain
40%	8	7	6	6	Total Trades
	4	6	5	6	Winning Trades
	50.00%	85.71%	83.33%	100.00%	Percent Winners
	3.674%	42.118%	61.677%	131.121%	Cumulative % Gain
	0.459%	6.017%	10.279%	21.854%	Average % Gain
30%	18	10	12	12	Total Trades
	14	15	8	11	Winning Trades
	77.78%	150.00%	66.67%	91.67%	Percent Winners
	59.155%	56.245%	91.682%	232.834%	Cumulative % Gain
	3.286%	5.625%	7.640%	19.403%	Average % Gain
20%	28	22	18	15	Total Trades
	19	14	12	12	Winning Trades
	67.86%	63.64%	66.67%	80.00%	Percent Winners
	46.840%	75.350%	125.062%	225.349%	Cumulative % Gain
	1.673%	3.425%	6.948%	15.023%	Average % Gain

First, what we *cannot* conclude, the percentage of daily New Lows to use as a signal is not clear; they all appear to work well with various trade-offs.

Nevertheless, the results demonstrate a conclusions we *can* discern from the data.

The first is that the longer the period a position is held after a "New Lows Climax," the greater the gains, as well as likelihood of profitability. Secondly, even readings as low as 20% daily New Lows show significance looking out one calendar quarter or longer.

2. A Signal for Periods of "Extreme Vulnerability"

We now present a very novel indicator utilizing daily New Lows and their *persisting resonance* in weeks and months after significant readings wherein very low readings of daily New Lows can be used to isolate periods of "*Extreme Vulnerability*."

Essentially, in the markets of the late 20-teens and early 2020s, wherein we typically see roughly 3000 big board shares trading daily, we have used this signal with respect to looking for "single-digit" daily New Lows in order to record the signal. Thus, to extend this out to a percentage of daily New Lows so as to make it *atemporal*, we use a number of 0.325% of daily New Lows. Using 0.325% daily New Lows makes the daily New Low readings of the past equate to single digit daily New Low readings of the current era.

The signal is very simple but with profound market implications as evidenced historically in Figure 3. The signal is simply, *if we have not witnessed a reading as low as 0.325% or lower in daily New Lows for any of the past 65 (roughly 1 calendar quarter, or 3 months) trading days, we stipulate that we have entered a period of "Extreme Vulnerability."*

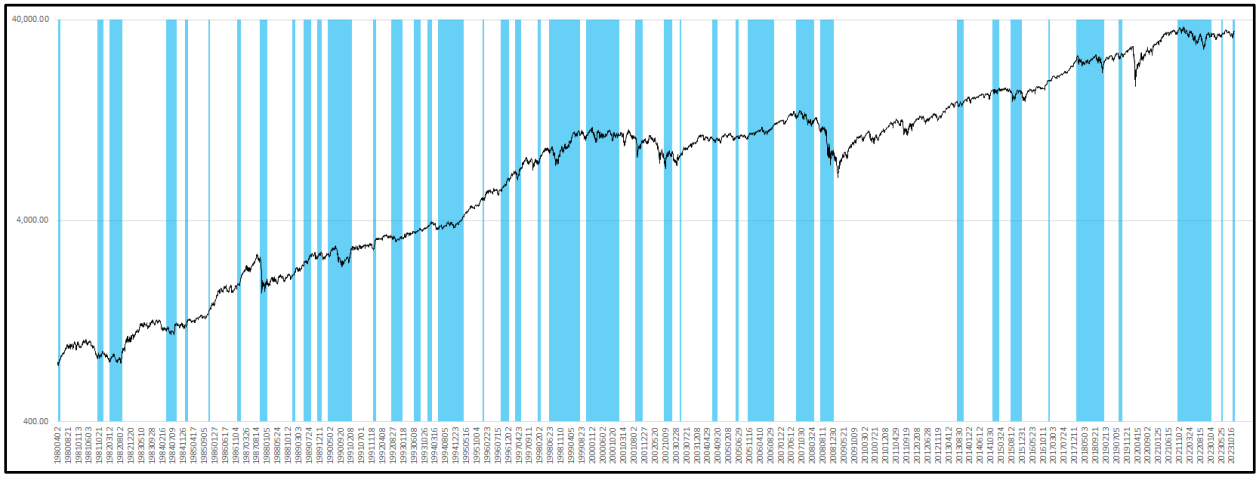


Figure 3: Lognormal Dow Industrials overlaid with periods of Extreme Vulnerability in blue.

Let us examine important market milestones of the past four-plus decades with respect to periods of Extreme Vulnerability.

Beginning with the Crash of October 1987, as displayed in Figure 4, the vulnerability appears on Sept 27, 1987.

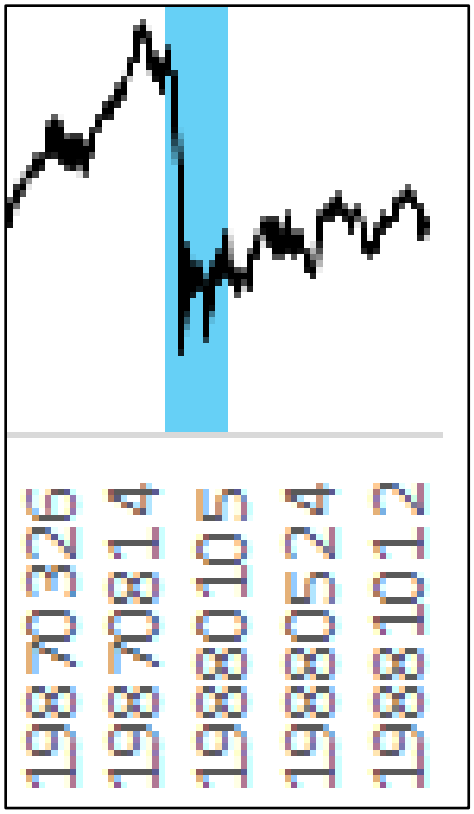


Figure 4: Lognormal Dow Industrials and the Crash of '87, transpired in a period of Extreme Vulnerability.

Next, we turn our attention to the Global Financial Crisis of 2007-9, as show in in Figure 5.

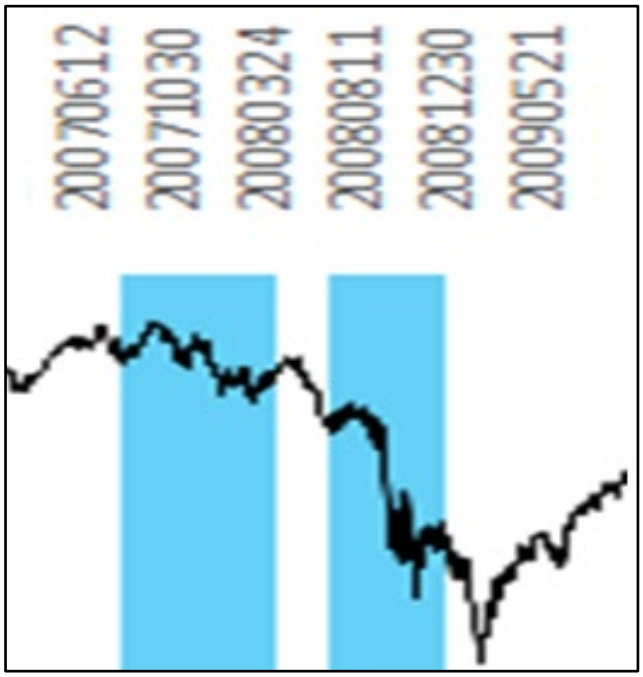


Figure 5: Lognormal Dow Industrials and the Global Financial Crisis of 2007-9.

Notice the period of Extreme Vulnerabiity, as with most of the major tops over this 44-year time-span, was flagged by this indicator (contrary to what one might expect, late stage bull moves are not characterized by days of extreme low readings of daily New Lows). There was a period of a few months in late Spring, early Summer of 2008 where things cleared (typical in a major bear market as evidenced by the history in Figure 3), as well as in the final stages of the bear market beginning late in 2008 where this signal subsided.

Turning our attention to the major moves of the most recent 6 of the 44 years under examination, shownin Figure 6, we can say we have seen four major market drops from significant tops.

Figure 6: Lognormal Dow Industrials from the Summer of 2017 to the present.

The "Volmageddon" incident of February 5, 2018, wherein President Trump's tweet about tariffs caused a very strong drop in stocks that day coupled with a meltdown in inverse volatility products that evening, occurred during a period of Extreme Vulnerability that went into effect during the third week of January, and remained in force into the fourth quarter where the rate panic began.

The onset of the Covid selloff in March of 2020 was not marked as a period of Extreme Vulnerability, one of the few major market drops of the 44-year time-span ensued which was not precipitated by this condition.

The top of late 2021 / early 2022 however, did transpire under a period of Extreme Vulnerability, a condition which noticeably did not appear until just before the top, steering clear of giving confusing signals to the analyst in the post-Covid run-up.

3. The "Bloodbath Sidestepping" 4% Rule

This is perhaps the most powerful and reliable of all rules we have seen pertaining to daily New Lows. The rule is, if we experience a day with more than 4% daily New Lows, we go short or flat (or fully-hedge our long position). A perfunctory glance, graphically, of this indicator over the past 44 years appears as Figure 7.

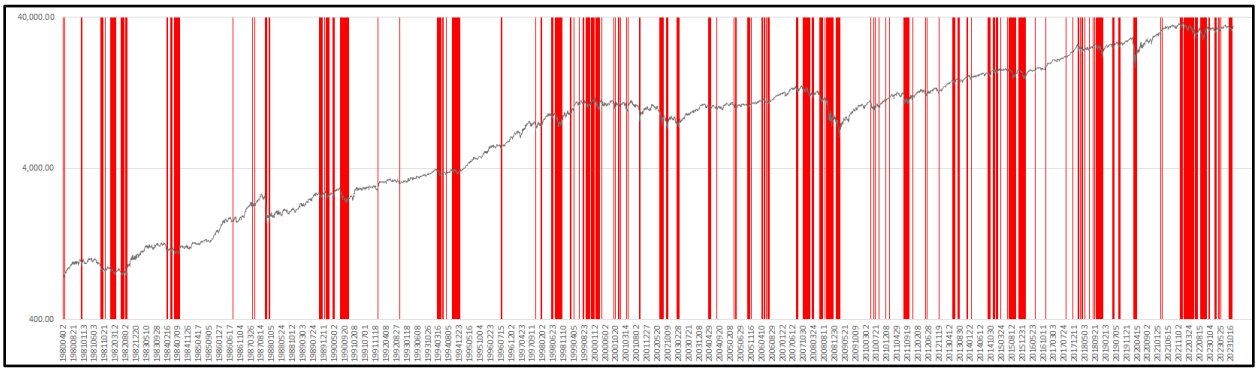


Figure 6: Lognormal Dow Industrials with days of 4% or more daily New Lows superimposed in red, January 2, 1980 – November 20, 2023.

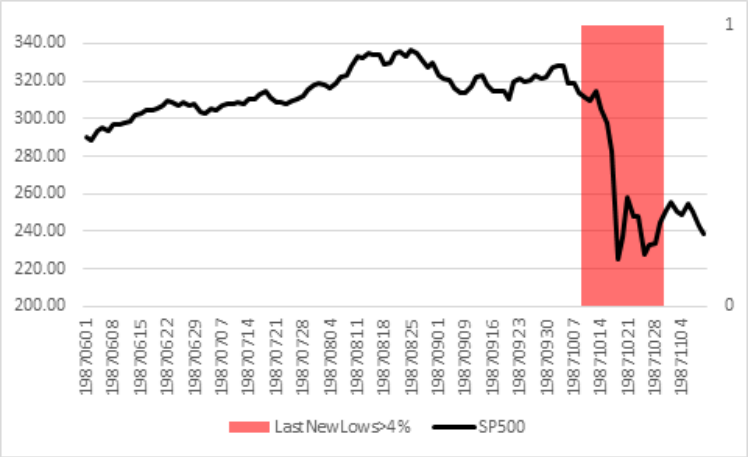
The indicator is far from perfect, but if we examine the hard-and-fast rule of going short, flat or fully hedged into the close of a day that is seeing 4% or more daily New Lows (and you know this value before the close of trading, as it is broadcasted throughout the trading day, and can only increase throughout the day), and re-establishing our position into the close on the first subsequent day where daily New Lows are less than 4%, we find the following track record.

We will examine the track record where we are always either long or short based on this indicator. First there were 630 trades of this signal of which only 257 were profitable (40.79% winning trades). As evidenced in Figure 7, **<u>however, nearly every major market selloff in the 44-year history saw an unusually high number of such days and the rule gets us out early on in such drops.</u>** As a result, over the nearly 44-year time-period, this simple "*bloodbath-sidestepping*" rule alone, despite its numerous, short-lived, false signals along the way, netted 10,981.86 Dow Jones Industrial Average points more than mere buy & hold over that period. That's greater than a 31 percent improvement by standards of today's DJI price (November 20, 2023 close being 35,151.04). This 10,981.96-point surfeit was garnered on an DJI that closed as low as 759.13 in the period under examination.

This "Bloodbath Indicator," based on daily readings, gives the analyst advance warnings that what he is witnessing has a very high chance of being more than a mere, prosaic market down day, but may well be the early stages of a market bloodbath. In fact, the analyst would be very hard pressed to find a market "bloodbath" condition that did not exhibit this signal!

We now examine a few critical points in this test period. Again, looking back at the October, 1987 Crash in Figure 7.

Figure 7: Lognormal S&P 500 with days of 4% or more daily New Lows superimposed in red, over the October '87 Crash.



October 9, a full week and a day before the Crash, we started to see this indicator serve up readings of greater than 4% daily New Lows. This indicator would have had one implementing it either short, flat or fully-hedged into this bloodbath.

We now turn our sights to the past six years, as we see in Figure 8.

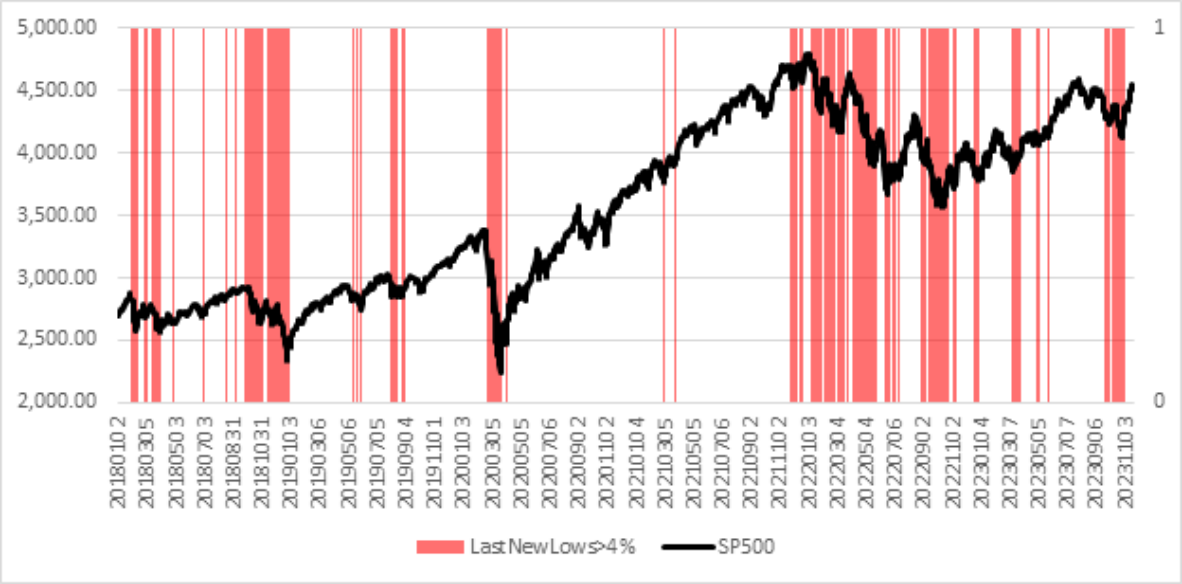


Figure 8: Lognormal S&P 500 with days of 4% or more daily New Lows superimposed in red, January 2018 through November 2023.

Let us zoom-in and examine once again the Volmageddon episode of February 2018, shown in Table 2.

DATE	SP500	%NL
20180124	2,837.54	1.27%
20180125	2,839.25	2.19%
20180126	2,872.87	2.66%
20180129	2,853.53	6.61%
20180130	2,822.43	8.19%
20180131	2,823.81	3.69%
20180201	2,821.98	4.24%
20180202	2,762.13	11.15%
20180205	2,648.94	14.61%
20180206	2,695.14	15.43%
20180207	2,681.66	1.48%
20180208	2,581.00	6.73%
20180209	2,619.55	11.56%
20180212	2,656.00	2.59%
20180213	2,662.94	1.78%
20180214	2,698.63	2.62%
20180215	2,731.20	1.55%

Table 2: Volmageddon and Percentage of New Lows.

The actual Volmageddon incident occurred on February 5. The hard-and-fast application of this rule would have had the implementor of this rule out or short on the close of February 2, or fully hedged by the close of that day, sidestepping Volmageddon entirely, perhaps even profiting off of it had they opted to be short for such signals.

Skipping ahead to the Covid "bloodbath," (the reader can readily see how the rate hikes of Q4 2018 were handled by this indicator), we see the results of this expressed in Table 3.

DATE	SP500	LastNewLow
20200219	3,386.15	0
20200220	3,373.23	0
20200221	3,337.75	0
20200224	3,225.89	1
20200225	3,128.21	1
20200226	3,116.39	1
20200227	2,978.76	1
20200228	2,954.22	1
20200302	3,090.23	1
20200303	3,003.37	1
20200304	3,130.12	1
20200305	3,023.94	1
20200306	2,972.37	1
20200309	2,746.56	1
20200310	2,882.23	1
20200311	2,741.38	1
20200312	2,480.64	1
20200313	2,711.02	1
20200316	2,386.13	1
20200317	2,529.19	1
20200318	2,398.10	1
20200319	2,409.39	1
20200320	2,304.92	1
20200323	2,237.40	1
20200324	2,447.33	0
20200325	2,475.56	0

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Here we clearly see the "Bloodbath Indicator," having sidestepped nearly the entirety of the sudden, Covid selloff, or having been short on it throughout the 21-day long period of greater than 4% daily New Lows.

Lastly, let us examine this rule on the full-history we possess of daily New Lows data. Despite the pre-1980 flaws, the robustness of the "Bloodbath Indicator" clearly holds up. This is presented as Figure 9.

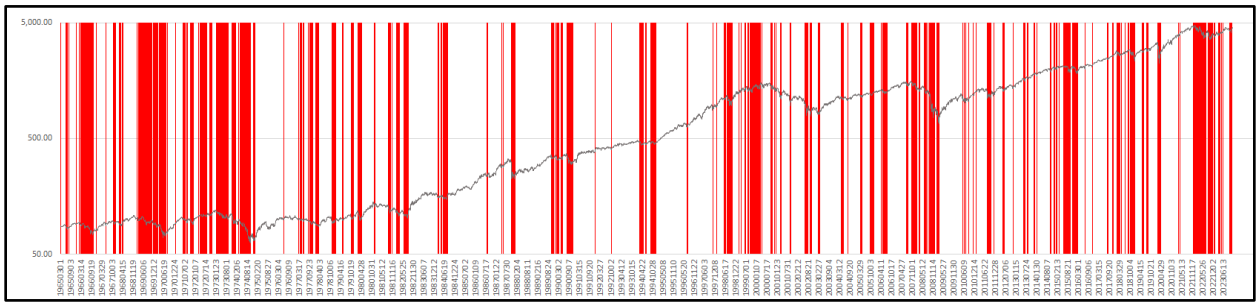

Figure 9: Lognormal S&P 500 with days of 4% or more daily New Lows superimposed in red, March 1, 1965 – November 20, 2023.

Summary and Conclusion

We have unequivocally demonstrated using empirical data, not only the value of daily perusal of New Lows to the analyst, but the necessity of such, given the reliability of these indicators based on the past four-plus decades of daily data.

It is important to note that, at no point, do we advocate using daily New Lows in isolation. Naturally, what we present should be used with the rest of the analyst's tools. Our point here is to demonstrate why the analyst should likely be paying more attention to daily New Lows, provide new tools to do this, and point out the "future resonance," that days of extreme readings in daily New Lows possess; this is not a data point to look at with respect to simply the latest daily reading.

It is exactly this "future resonance," the ripples in subsequent weeks and months that single, extreme daily readings in this sole data element signal to the analyst

Thus, the central and *novel* insight of this paper:

> Days of extreme readings, either high or low, in daily New Lows, create an "echo" much like waves in a pond resonating out from a stone thrown into it. Such days have been shown to persist in terms of their effect on price action for several months afterwards.

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